

**GABAY: A HIERARCHICAL MULTI-AGENT RETRIEVAL-AUGMENTED
CONVERSATIONAL AI SYSTEM FOR COLLEGE MANAGEMENT AND
DECISION SUPPORT**

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This thesis proposal, **GABAY: A Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System for College Management and Decision Support**, prepared and submitted by **Jayvee G. Reyes, Jamaica Fea C. Aquino, Johren Mae B. Biscarra, Rica U. Castronuevo, and Karylle P. Montemayor**, in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science, was examined and passed on September 3, 2026, by the Thesis Committee composed of:

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JR
JFA
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DEDICATION

*This study is wholeheartedly dedicated to our **PARENTS**, whose unconditional love, continuous moral guidance, and unwavering financial support served as our pillar of strength throughout this journey. Thank you for your tireless sacrifices, for meeting our needs during the development of this work, and for teaching us that even the most daunting endeavors can be achieved through perseverance, patience, and taking things one step at a time.*

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ABSTRACT

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Large language models and Retrieval-Augmented Generation (RAG) systems have improved information retrieval and conversational assistance; however, existing systems are commonly designed for specific applications and do not provide an integrated approach for retrieving institution-specific knowledge, coordinating multiple academic and administrative domains, and supporting college-level decision-making.

To address this gap, this study proposes GABAY: A Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System for College Management and Decision Support for the College of Computer Science of Don Mariano Marcos Memorial State University–South La Union Campus. The study aims to design a hierarchical multi-agent RAG architecture and develop GABAY to support institutional knowledge retrieval, college management, decision support, and report generation.

A developmental-experimental research design will be employed, combining systematic system development with controlled testing of the proposed architecture. The technical performance of GABAY will be evaluated in terms of retrieval accuracy, response quality, reasoning capability, and report generation accuracy, while its technology acceptance will be determined among administrators, faculty and staff, and students using an extended Technology Acceptance Model (TAM). The study is expected to provide an intelligent and integrated conversational platform that improves access to institutional information and supports efficient, evidence-based academic and administrative decision-making.

Keywords: *conversational AI system, hierarchical multi-agent, retrieval-augmented generation*

Chapter 1

INTRODUCTION

Situation Analysis

Large language models (LLMs) have become powerful tools for answering questions, writing documents, and carrying on conversations. Despite these strengths, they still have important limitations. They can produce incorrect information, rely on outdated knowledge, and cannot access institution-specific data unless that information is provided during the conversation. Retrieval-Augmented Generation (RAG) addresses these challenges by retrieving relevant documents from an external knowledge base and using them to support the model's responses. As RAG technology has evolved, researchers have moved from simple single-agent designs to more advanced hierarchical multi-agent architectures. This progression provides the foundation for developing a Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System designed for the College of Computer Science at Don Mariano Marcos Memorial State University–South La Union Campus (DMMMSU-SLUC).

Retrieval-Augmented Generation (RAG) has emerged as a significant advancement in conversational artificial intelligence by improving the reliability and factual accuracy of large language models (LLMs). Unlike conventional LLMs that rely solely on knowledge acquired during pre-training, RAG retrieves relevant information from external knowledge sources before generating a response. This retrieval process enables language models to produce answers that are grounded in verified documents, thereby reducing hallucinations and improving access to current and domain-specific information (Sharma, 2025; Asai et

al., 2024). Nevertheless, existing RAG systems continue to encounter challenges related to irrelevant document retrieval, scalability, and complex multi-step reasoning, which have driven the continuous evolution of RAG architectures (Oche et al., 2025).

The earliest implementation, commonly referred to as Traditional Retrieval-Augmented Generation, follows a fixed retrieval pipeline. A user's query is embedded into a vector representation, matched against a vector database, and supplemented with retrieved documents before the language model generates a response (Sharma, 2025). This architecture effectively improves factual consistency because responses are supported by external knowledge rather than relying entirely on the model's internal parameters. However, its rigid workflow limits its ability to revise retrieval strategies or perform additional searches when the initially retrieved documents are insufficient. Consequently, Traditional RAG often struggles with questions that require reasoning across multiple documents or iterative information retrieval, highlighting the need for more adaptive retrieval approaches (Oche et al., 2025).

To overcome these limitations, researchers introduced Agentic Retrieval-Augmented Generation (Agentic RAG), which extends the traditional retrieval pipeline by incorporating autonomous decision-making into the retrieval process. Rather than relying on a single retrieval cycle, Agentic RAG enables intelligent agents to determine when additional retrieval is necessary, evaluate the relevance of retrieved information, and refine the search strategy before generating a final response (Oche et al., 2025). This adaptive capability allows the system to better handle complex and knowledge-intensive tasks that require multiple stages of reasoning.

Several studies have demonstrated the advantages of Agentic RAG. Self-RAG enables the language model to determine whether external retrieval is necessary and evaluate whether the retrieved documents adequately support the generated response, improving factual accuracy and response reliability (Asai et al., 2024). Likewise, Reflective RAG introduces continuous self-evaluation throughout the retrieval and reasoning process, outperforming baseline Agentic RAG systems across multiple knowledge-intensive question-answering benchmarks (Wu et al., 2025). Speculative RAG further improves computational efficiency by employing a smaller language model to guide retrieval before invoking a larger model, reducing inference latency while maintaining high reasoning performance (Wang et al., 2024). Collectively, these developments demonstrate that RAG has evolved from a static retrieval mechanism into a more adaptive and intelligent framework. However, despite these advances, most Agentic RAG implementations still depend on a single language model to perform retrieval, reasoning, and response generation, creating new challenges as applications become increasingly complex.

The limitations of single-model reasoning led to the development of single-agent and eventually multi-agent architectures. In a single-agent architecture, one language model performs all major tasks, including intent recognition, document retrieval, reasoning, and response generation. This centralized design is suitable for straightforward applications because all decisions are handled by a single intelligent agent. However, assigning all responsibilities to one model limits the system's ability to process multiple tasks simultaneously, integrate information from different knowledge sources, and

maintain consistent performance when solving complex institutional problems (Buller et al., 2024; Mendoza et al., 2025).

Studies in academic environments demonstrate both the strengths and limitations of single-agent systems. A GPT-4-based academic advising system developed at the United States Military Academy achieved high accuracy in answering institutional policy questions but struggled to reason across multiple handbook sections (Buller et al., 2024). Similarly, a GPT-4o academic advising chatbot implemented at the New York City College of Technology provided clear and policy-consistent responses but remained susceptible to structural hallucinations when integrating information from multiple institutional documents (Mendoza et al., 2025). Aurora, a neuro-symbolic advising system developed at Florida International University, improved response efficiency by combining GPT-4o-mini with a rule-based reasoning engine; however, its dependence on predefined database schemas limited its ability to process unstructured academic scenarios (Quincoso Lugones et al., 2025). These studies collectively suggest that although single-agent architectures perform well in narrowly defined applications, they become less effective as institutional tasks require broader reasoning and coordination.

To improve task specialization, researchers introduced role-based multi-agent architectures, in which several intelligent agents collaborate by performing specialized responsibilities such as retrieval, reasoning, verification, and response generation (Fang et al., 2026; Nguyen et al., 2026). Compared with single-agent systems, this division of labor enables more effective handling of complex queries because each agent focuses on a specific component of the reasoning process. Studies such as Orion, MA-RAG, university admissions counseling systems, and AdvisorX demonstrated improvements in retrieval

accuracy, reasoning capability, and response quality through collaborative multi-agent processing (Fang et al., 2026; Jiang et al., 2026; Nguyen et al., 2026; Nguyen-Duc et al., 2026). Despite these improvements, collaboration among multiple agents also introduces coordination challenges. Communication failures, task misalignment, and cascading errors remain common obstacles that reduce system reliability in real-world deployments (Nechepurenko & Shuvalov, 2026; Xie et al., 2026).

To address these coordination issues, researchers proposed hierarchical multi-agent architectures, where a high-level supervisor agent coordinates multiple specialized worker agents responsible for different subtasks (Park & Kwon, 2026; Liu et al., 2026). This layered architecture combines centralized task management with distributed expertise, enabling more efficient task decomposition, coordinated reasoning, and improved decision-making. Recent studies have shown that hierarchical architectures outperform earlier approaches in reasoning capability, coordination efficiency, and retrieval effectiveness (Liu et al., 2026; Park & Kwon, 2026; Torkian & Zhou, 2024). Although routing errors and information loss between hierarchical layers remain existing challenges, these architectures represent the most comprehensive approach currently available for managing complex institutional tasks. Their ability to coordinate specialized agents while maintaining a global view of the overall problem makes them particularly suitable for higher education environments that require integrated academic management, decision support, and learning services.

Despite these advances, higher education institutions continue to face challenges in managing and accessing institutional knowledge efficiently. The College of Computer Science (CCS) at Don Mariano Marcos Memorial State University–South La Union

Campus (DMMMSU-SLUC) is no exception. The College manages various academic and administrative functions, including student advising, curriculum management, enrollment, prerequisite verification, research and extension coordination, quality assurance, student services, and institutional reporting. These functions rely on institutional resources such as student handbooks, curriculum guides, university memoranda, course catalogs, accreditation reports, administrative policies, and operational databases. These information resources originate from multiple institutional systems and repositories, requiring users to consult several independent sources before obtaining the information needed for academic or administrative decision-making. Consequently, information retrieval, report preparation, and decision-making often involve manual searches across multiple documents and systems, making these processes both time-consuming and inefficient.

Students frequently seek information regarding enrollment procedures, curriculum requirements, course prerequisites, scholarships, academic policies, and graduation requirements. Likewise, faculty members and administrators regularly verify academic records, interpret university policies, prepare accreditation documents, monitor research and extension accomplishments, generate institutional reports, analyze academic performance, and support institutional planning and decision-making. These activities require information from multiple institutional documents and databases, making the process both time-consuming and labor-intensive.

At present, these tasks are commonly performed through face-to-face consultations, email exchanges, messaging platforms, spreadsheet-based reports, and manual review of institutional documents. Preparing reports, summarizing institutional data, and generating charts, dashboards, and visual presentations for planning and

accreditation also require significant manual effort. While these approaches remain functional, they become increasingly inefficient during enrollment periods, accreditation activities, and other peak academic operations, resulting in longer response times, increased administrative workload, and inconsistencies in both information retrieval and institutional reporting.

Another major challenge is the lack of an intelligent conversational system that can retrieve and validate information directly from official university documents. Existing communication methods rely heavily on manual interpretation of institutional documents, making response accuracy and consistency dependent on the knowledge and experience of the personnel providing assistance. Without a retrieval mechanism that grounds responses in verified institutional documents, there is a greater risk of incomplete, outdated, or inconsistent academic guidance. As previous studies emphasize, conversational AI systems without reliable retrieval mechanisms remain vulnerable to hallucinations and factual inaccuracies, particularly in knowledge-intensive environments where institutional information changes over time (Asai et al., 2024; Oche et al., 2025).

These challenges show that the College needs more than a conventional chatbot that answers frequently asked questions. Instead, the College needs an intelligent conversational system that can retrieve and integrate information from official university knowledge sources, reason across multiple institutional documents, generate reports and visual summaries, and support data-driven decision-making for academic and administrative operations. Such a system should also provide accurate, consistent, and context-aware assistance to students, faculty members, administrators, and other stakeholders while improving information accessibility, operational efficiency, and

evidence-based institutional planning. Despite recent advances in Retrieval-Augmented Generation and multi-agent conversational AI, it remains unclear whether existing approaches can fully meet these institutional requirements. This unresolved challenge highlights the need for a solution specifically designed to meet the diverse academic and administrative needs of higher education institutions.

The reviewed literature demonstrates that Retrieval-Augmented Generation (RAG) and agent-based conversational AI have significantly improved information retrieval, reasoning, and response generation. Traditional RAG, Agentic RAG, single-agent, role-based multi-agent, and hierarchical multi-agent architectures have each addressed specific limitations of earlier approaches by enhancing factual accuracy, task specialization, and coordination (Asai et al., 2024; Buller et al., 2024; Fang et al., 2026; Liu et al., 2026; Oche et al., 2025; Park & Kwon, 2026). However, these systems are primarily designed for specific applications, such as academic advising, policy assistance, or domain-specific question answering, rather than providing a unified platform for managing the diverse academic and administrative functions of higher education institutions.

For the College of Computer Science at DMMMSU-SLUC, existing approaches do not fully address the need for an integrated conversational AI system that can retrieve and reason across institution-specific knowledge sources while supporting college management, decision support, learning services, institutional reporting, and other academic and administrative operations. Therefore, a hierarchical multi-agent Retrieval-Augmented Generation framework is needed to integrate college management, decision support, learning services, institutional reporting, and document-grounded reasoning

within a single intelligent platform tailored to the operational requirements of higher education institutions.

To address the identified research gap, this study proposes GABAY: A Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System for College Management, Decision Support, and Learning Services. The proposed system integrates Retrieval-Augmented Generation with a hierarchical multi-agent architecture to provide accurate, context-aware, and explainable responses using official institutional knowledge sources. Unlike existing conversational AI systems that primarily focus on individual applications, GABAY is designed as a unified platform that supports multiple academic and administrative functions within the College of Computer Science at Don Mariano Marcos Memorial State University–South La Union Campus (DMMMSU-SLUC).

The system is designed to assist students, faculty members, administrators, and other stakeholders by retrieving and integrating information from institutional knowledge sources, generating reports and visual summaries, supporting decision support activities, and assisting academic and administrative operations through coordinated reasoning among specialized intelligent agents. Through its hierarchical coordination of specialized intelligent agents, GABAY aims to improve information accessibility, operational efficiency, institutional reporting, and the consistency of academic and administrative services while reducing the manual effort required in routine institutional processes.

Beyond addressing DMMMSU-SLUC's operational needs, GABAY advances conversational artificial intelligence in higher education by showing how Retrieval-Augmented Generation and hierarchical multi-agent architectures can be integrated into a single intelligent platform for college management, decision support, and learning services.

The proposed system also supports the University's digital transformation initiatives while contributing to Sustainable Development Goal (SDG) 4: Quality Education, SDG 9: Industry, Innovation and Infrastructure, SDG 10: Reduced Inequalities, and SDG 16: Peace, Justice and Strong Institutions through improved access to institutional knowledge, data-driven decision-making, and transparent academic governance.

Statement of Objectives

The general objective of this study is to design, develop, and evaluate GABAY, a Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System that supports college management and decision support through intelligent retrieval, reasoning, and generation of institution-specific information for the College of Computer Science at Don Mariano Marcos Memorial State University–South La Union Campus (DMMMSU-SLUC).

Specifically, this study aims to:

1. Design a hierarchical multi-agent Retrieval-Augmented Generation architecture for institutional knowledge retrieval and conversational reasoning;
2. Develop GABAY to support college management and decision support through institutional knowledge retrieval and report generation.
3. Evaluate the performance of GABAY in terms of:
 - 3.1 Retrieval accuracy;
 - 3.2 Response quality;
 - 3.3 Reasoning capability; and
 - 3.4 Report generation accuracy

4. Determine the technology acceptance of GABAY among administrators, faculty and staff, and students using an extended Technology Acceptance Model (TAM) in terms of:
 - 4.1 Perceived usefulness;
 - 4.2 Perceived ease of use;
 - 4.3 Trust in the AI system; and
 - 4.4 Behavioral intention to use

Time and Place of the Study

The study will be conducted from June 2026 to March 2027 within the premises of the Don Mariano Marcos Memorial State University – South La Union Campus (DMMMSU-SLUC) in Agoo, La Union, specifically in the College of Computer Science (CCS).

Definition of Terms

For a better understanding of this research report, the following terms are operationally defined:

Behavioral intention to use refers to the willingness of students, faculty members, non-teaching staff, and administrators of DMMMSU-SLUC CCS to use GABAY for their academic and administrative tasks.

College Management refers to the academic and administrative activities supported by GABAY, such as student advising, curriculum and prerequisite checking, research and extension monitoring, quality assurance, and preparation of institutional reports.

Conversational AI refers to the natural language feature of GABAY that allows users to ask questions and receive answers based on available institutional documents and data.

Conversational reasoning refers to the process used by GABAY to understand a user's question, identify what information is needed, and use the retrieved information to form an appropriate response.

Decision Support refers to the ability of GABAY to provide useful information, summaries, reports, and data visualizations that can help faculty members and administrators make informed academic and administrative decisions.

Document summarization refers to the ability of GABAY to identify the important information from institutional documents and present it in a shorter and clearer form while keeping the main ideas and essential details.

Gabay refers to the Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System to be developed in this study for the College of Computer Science of DMMMSU-SLUC. It supports institutional knowledge retrieval, college management, and decision-making.

Hierarchical Multi-Agent RAG is Gabay's system architecture, in which a coordinator agent manages specialized agents that perform specific tasks. These agents work together to retrieve information, process user requests, apply relevant rules or policies, and generate responses.

Institutional knowledge retrieval is the process of finding relevant information from official institutional sources, such as student handbooks, curricula, memoranda, policy manuals, and other authorized documents and records.

Institutional Report Generation refers to the ability of GABAY to collect relevant institutional data and automatically organize it into structured summaries, tables, and reports needed for academic and administrative purposes.

Perceived ease of use refers to the degree to which users believe that GABAY is clear, easy to learn, and easy to use.

Perceived usefulness refers to the degree to which users believe that GABAY can help them perform academic and administrative tasks more quickly and efficiently.

Performance of Gabay refers to how well the system performs its intended technical tasks. In this study, it will be evaluated in terms of retrieval accuracy, response quality, reasoning capability, and report generation accuracy.

Reasoning capability refers to how well GABAY performs tasks that require multiple steps, such as applying rules, connecting information from different sources, and arriving at the correct response.

Report generation accuracy refers to the degree to which the information in institutional reports generated by GABAY matches the correct or expected institutional data.

Response quality refers to how well GABAY answers a user's question based on the retrieved information. It considers whether the response is accurate, relevant, complete, clear, and supported by the available institutional information.

Retrieval accuracy refers to GABAY's ability to find correct, relevant information from institutional documents and data sources based on a user's request.

Retrieval-Augmented Generation refers to the process GABAY will use to retrieve relevant information from institutional documents or data sources before the AI generates

a response. This helps the system produce answers that are supported by available institutional information.

Technology Acceptance Model refers to the framework that will be used in this study to determine users' acceptance of GABAY based on perceived usefulness, perceived ease of use, trust in the AI system, and behavioral intention to use.

Trust in the AI system refers to the degree to which users believe that GABAY provides reliable and accurate information, follows institutional policies, and handles institutional data appropriately.

Visualization of Operational Data refers to the ability of GABAY to present institutional data using charts, graphs, and other visual formats to help users understand information and support decision-making.

Chapter 2

METHODOLOGY

Research Design

This study will employ a developmental-experimental research design, which combines the systematic development of a system with controlled experiments to design, test, and evaluate its technical components. This design is appropriate because it allows the researchers to develop GABAY as a complete system while conducting experiments on its core Hierarchical Multi-Agent Retrieval-Augmented Generation (RAG) architecture.

The developmental phase refers to the systematic process of designing, developing, testing, and refining a system to meet its intended purpose and user requirements (Richey & Klein, 2005). In this study, this phase will cover the requirements analysis, system design, development, integration, testing, and refinement of GABAY for college management and decision support.

The experimental phase involves controlled testing of different conditions or configurations to determine their effects on specific outcomes (Creswell & Creswell, 2018). In this study, this phase will involve controlled testing and refinement of the Hierarchical Multi-Agent RAG architecture using predefined benchmark queries and controlled system configurations. The researchers will examine the retrieval process, agent routing, multi-step tool execution, and response generation under standardized test conditions. The resulting architecture will then be evaluated using predefined technical performance measures.

Materials and Procedures

Figure 1 presents the block diagram of the study's procedures carried out by the researchers in achieving the objectives posed in this study.

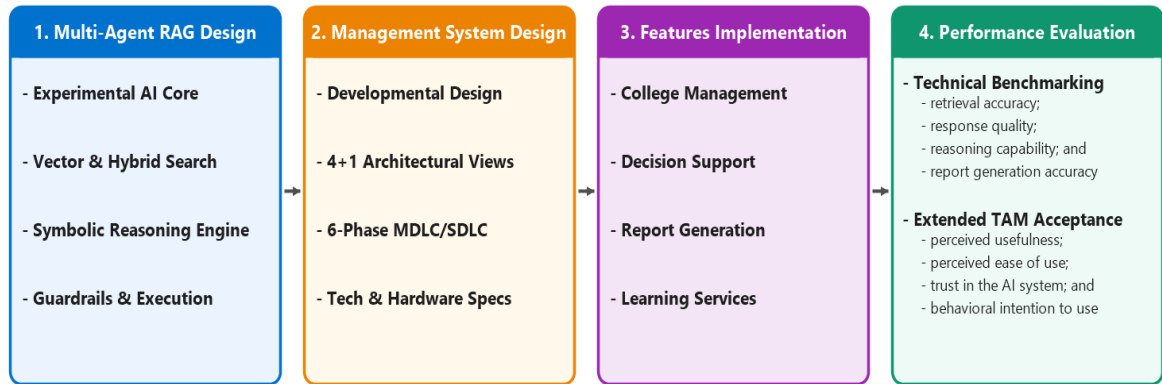


Fig.1. Block Diagram of Procedures in Achieving Study's Objectives

Design of Hierarchical Multi-Agent RAG Architecture

In line with Objective No. 1, the design of the Hierarchical Multi-Agent Retrieval-Augmented Generation (RAG) Architecture will serve as the core experimental component of this study. A Hierarchical Multi-Agent RAG architecture will be designed to support institutional knowledge retrieval, conversational reasoning, college management, and decision support within the College of Computer Science of DMMMSU-SLUC. The proposed architecture will implement retrieval, reasoning, agent coordination, privacy enforcement, and response generation mechanisms to provide users with relevant and evidence-based responses. As shown in Figure 2, the architecture will consist of seven major components: Institutional Data Collections, Upstream Ingestion, Qdrant Vector Store, Retrieval Pipeline, Coordinator Swarm, Actuation Assembly, and User Chat Query.

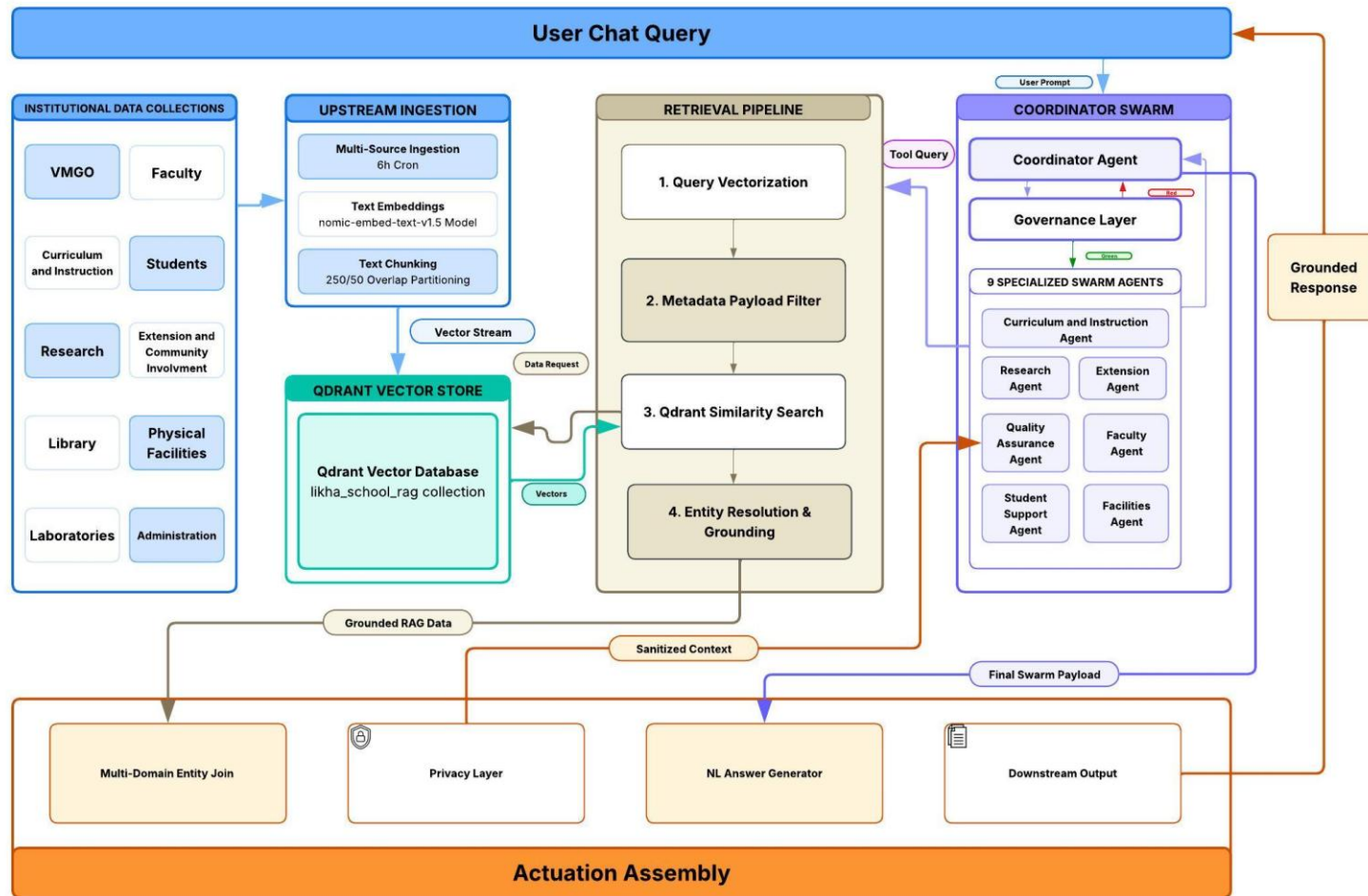


Fig.2. Hierarchical Multi-Agent RAG Architecture Framework

Institutional Data Collections. This serves as the centralized repository of academic, administrative, and operational information that supports organizational processes and decision-making (Tiwari et al., 2026). In this study, institutional data will be collected from authorized sources, including curriculum and instruction, research, extension, quality assurance, faculty, support to students, and facilities through Likha ERP. These data sources will serve as the primary knowledge base of GABAY.

Upstream Ingestion. This refers to the process of extracting, transforming, and preparing data before they are stored and used for retrieval and reasoning tasks (Gao et al., 2023; Nussbaum & Nomic AI Team, 2024). In GABAY, institutional data from Likha ERP will be processed through an automated Extract, Transform, and Load (ETL) pipeline. The extracted text is divided into manageable portions using a Recursive Character Text Splitter configured with a 250-character chunk size and a 50-character overlap (250/50). This slight overlap ensures that complete thoughts and sentences are not cut off at chunk boundaries (Asai et al., 2024; Gao et al., 2023). These text chunks are then converted into numerical representations, known as 768-dimensional vector embeddings, using the open-source nomic-embed-text-v1.5 model, enabling the AI to understand the semantic meaning of the text (Nussbaum et al., 2024). Each chunk is also tagged with descriptive labels (metadata) that identify its original document source and college department (such as Research, Extension, QA, etc.). Finally, these vector representations and their metadata tags are sent to the Qdrant Vector Store, which functions as a specialized search database that organizes, stores, and rapidly retrieves matching information when a user asks a question.

Qdrant Vector Store. The Qdrant Vector Store is a vector database designed to store, index, and retrieve vector embeddings together with their metadata (Oche et al.,

2025; Wang et al., 2024). Following the Upstream Ingestion process, the generated vector embeddings and their metadata will be stored in the dedicated `likha_school_rag` collection in Qdrant. Metadata lineage tags and domain attributes (`searchFilterJson: metadata.domain`) will be attached to each stored vector to identify its source and institutional domain. During query processing, the Qdrant Vector Store will receive requests from the Retrieval Pipeline and return the most relevant vector data based on similarity search and metadata filtering. In this way, Qdrant will serve as the main vector repository that will connect the upstream ingestion process with the retrieval of relevant institutional information.

Retrieval Pipeline. This will be responsible for identifying and retrieving information relevant to a user's query from the Qdrant Vector Store (Asai et al., 2024; Oche et al., 2025). Following the storage of vector embeddings in Qdrant, GABAY will use Dense (Vector) Retrieval as its core retrieval approach. The retrieval process will consist of four stages. Stage 1 (Query Vectorization) will convert the query into a 768-dimensional vector to enable the AI to understand its semantic meaning. Stage 2 (Metadata Payload Filtering) will apply domain filtering to limit the search to the appropriate institutional domain and reduce unrelated results. Stage 3 (Qdrant Similarity Search) will use cosine similarity to compare the query vector with the stored vectors and retrieve the top eight most relevant results (Asai et al., 2024; Gao et al., 2023). Finally, Stage 4 (Entity Resolution and Grounding) will link the retrieved information to its corresponding verified institutional source. The resulting information will then be forwarded to the appropriate component of the architecture for further processing and response generation.

Coordinator & Swarm. This will represent the hierarchical multi-agent component of the proposed architecture, where a Coordinator Agent will manage and coordinate specialized agents (Nechepurenko & Shuvalov, 2026; Park & Kwon, 2026; Silvestri et al., 2024). In GABAY, the Coordinator Agent, powered by GPT-4o-mini, will receive and analyze user queries to determine their scope and the appropriate domain for processing. The query will then pass through the Governance Layer, which will apply a traffic-light policy control mechanism based on the noise-redefinition approach described by Silvestri et al. (2024). Queries classified as Red due to unclear content or policy violations will be returned to the Coordinator Agent for appropriate handling, while queries classified as Green will be forwarded to the appropriate specialized agent.

The architecture will include seven specialized agents: Curriculum and Instruction agent, Research agent, Extension agent, Quality Assurance agent, Faculty agent, Student Support Agent, and Facilities agent, each operating using GPT-4o-mini. The selected agent will communicate with the Retrieval Pipeline to obtain relevant institutional information needed to process the query. The retrieved information will pass through the appropriate privacy controls before being provided to the agent. The specialized agent will then process the information and return its result to the Coordinator Agent for further processing and response generation.

Actuation Assembly. This refers to the component responsible for integrating retrieved information, protecting sensitive data, and preparing the final response for the user (Nguyen-Duc et al., 2025; Singh et al., 2026). Following Entity Resolution and Grounding, the retrieved RAG data will enter the Multi-Domain Entity Join, where related information from academic, administrative, and other institutional domains will be

combined when needed. The combined information will then pass through the Privacy Layer, where sensitive information will be anonymized or protected in accordance with Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012. The resulting sanitized information will then be returned to the appropriate agents for further processing.

After the Coordinator Agent receives and processes the validated results from the specialized agents, the final output will be forwarded to the Natural Language (NL) Answer Generator, which will transform the processed information into a clear and coherent natural language response supported by retrieved institutional information (Aftab et al., 2019; Nguyen-Duc et al., 2025). The generated response will then be sent to the Downstream Output, where it will be formatted and transmitted back to the GABAY platform for presentation to the user.

User Chat Query. The User Chat Query represents the primary interaction point between users and GABAY (Aftab et al., 2019). Administrators, faculty members, non-teaching staff, and students will submit natural language queries through the system's chat interface. The submitted query will pass through the different components of the Hierarchical Multi-Agent RAG architecture for retrieval, reasoning, and response generation. The final response generated by the system will then be presented to the user through the same interface.

Development of GABAY Management System

For Objective No. 2, which focuses on developing the GABAY Management System, this study will adopt the Research and Development (R&D) Cycle Framework, a structured methodology that combines rapid software prototyping with continuous, iterative refinement (Pookduang & Chansanam, 2026). This approach is appropriate for

developing GABAY because it supports systematic integration with Likha ERP while allowing researchers to progressively develop and refine its AI reasoning, information retrieval, system functionality, and security mechanisms without disrupting the existing institutional database (Pookduang & Chansanam, 2026).

As illustrated in Figure 3, the development process will consist of six iterative phases: (1) Problem Analysis and Requirements Specification, (2) System Architecture Design, (3) Prototype Development, (4) System Integration and Deployment, (5) Iterative Testing and Refinement, and (6) Multidimensional Evaluation. Although presented sequentially, these phases will follow an iterative process in which findings from testing and evaluation may be used to refine earlier development activities and improve the overall performance and usability of GABAY.

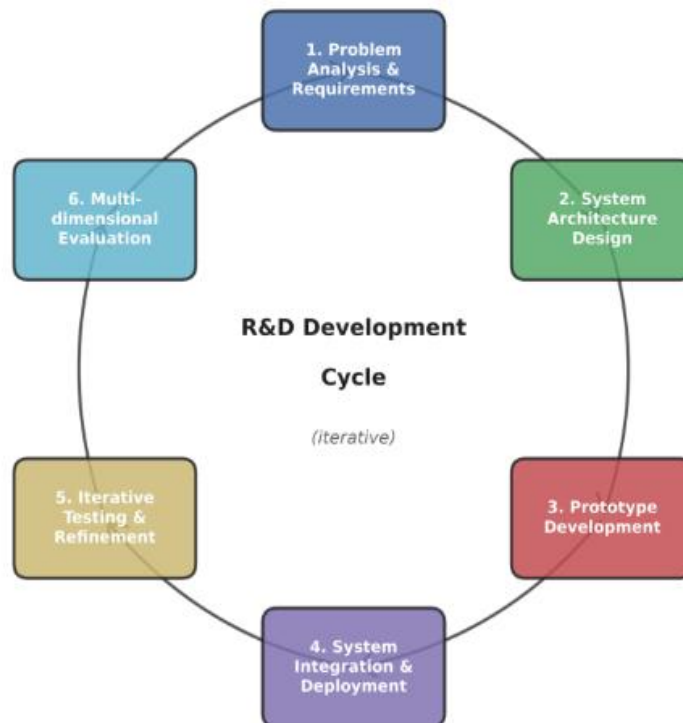


Fig.3. Research and Development Cycle Framework

Problem Analysis & Requirement Specification. This phase will establish the foundation for the development of GABAY by examining existing workflows, identifying operational challenges, and defining the functional and non-functional requirements of the system (Pookduang & Chansanam, 2026).

During this phase, the researchers will conduct interviews and document analysis with relevant DMMMSU-SLUC CCS stakeholders to understand existing academic and administrative processes, information needs, and challenges in accessing institutional information and preparing reports. The findings will be analyzed and translated into system requirements that will guide the development of GABAY. Functional requirements will include capabilities related to intelligent agent routing, retrieval of institutional information, document summarization, generation of reports and visual summaries, decision support, and academic assistance. Non-functional requirements will also be established, particularly for system performance, data privacy and security, usability, reliability, and concurrent user access.

System Architecture Design. This phase will translate the identified system requirements into a structured technical architecture that defines the major components of GABAY and how they will interact (Pookduang & Chansanam, 2026).

Based on the identified requirements, the researchers will design a decoupled API-driven modular architecture composed of three interconnected modules, as illustrated in Figure 4.

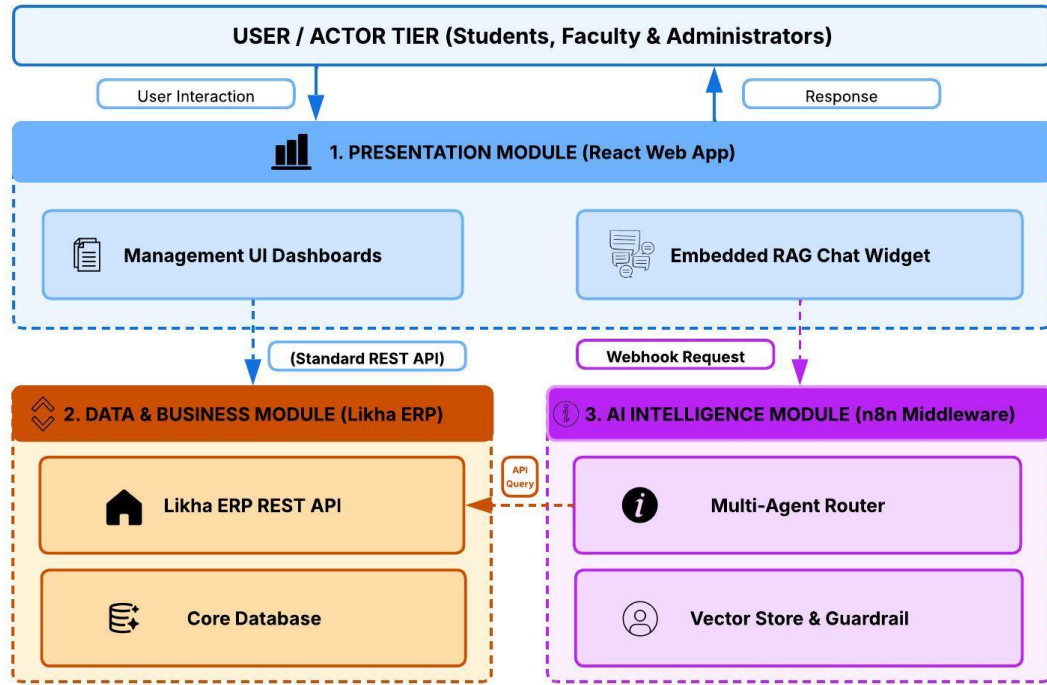


Fig.4. Decoupled API-Driven Modular Architecture Framework

The Presentation Module, developed as a React web application, will provide the management dashboards and embedded RAG chat interface for users. The Data and Business Module, supported by Likha ERP, will provide access to institutional data through its REST API while maintaining the core database. The AI Intelligence Module, implemented through n8n middleware, will handle AI-related processes, including multi-agent routing, retrieval from the vector store, and applying system guardrails. Communication among these modules will be established through standardized REST API and webhook connections, allowing the components to operate independently while maintaining controlled and secure data exchange (Pokala, 2021; Tiwari et al., 2026).

Since the Hierarchical Multi-Agent RAG Architecture will be designed under Objective No. 1, this phase will focus on integrating the proposed architecture into the

overall GABAY system design. The researchers will develop sequence diagrams to illustrate the flow of user requests and interactions among the system components, and a database design to define the data structures and relationships required by the system. User interface designs and wireframes will also be prepared to establish the layout, navigation, dashboards, chat interface, and other user-facing components of GABAY. These design artifacts will serve as the technical blueprint for the subsequent prototype development and system integration phases.

Prototype Development. This phase will focus on transforming the system architecture and design specifications into functional software components of the GABAY Management System (Pookduang & Chansanam, 2026).

Based on the technical designs established in the previous phase, the researchers will develop the core components of GABAY. This will include creating the responsive web interface using React Vite, implementing the management dashboards and embedded RAG chat interface, configuring n8n workflows for AI processing, and establishing the required API structures for communication with Likha ERP. The Hierarchical Multi-Agent RAG Architecture developed under Objective No. 1 will also be implemented by configuring the coordinator and specialized domain agents, retrieval mechanisms, vector store, and system guardrails within the prototype (Park & Kwon, 2026).

The developed components will undergo unit testing to verify their individual functionality before system-wide integration. Testing will cover key functions such as user interface interactions, API requests and responses, query routing, information retrieval, agent processing, and data formatting. Identified issues will be corrected at the component

level to ensure that the individual modules are functional and ready for integration in the succeeding phase.

System Integration and Deployment. This phase will focus on combining the independently developed components of GABAY into a unified and operational system and deploying the integrated application in a secure environment (Aftab et al., 2019).

Following the completion of the prototype, the researchers will integrate the React-based user interface, n8n AI workflows, Hierarchical Multi-Agent RAG Architecture, and Likha ERP backend through authenticated Directus and REST APIs. The integration will enable the different system components to exchange data and process user requests as a complete workflow. The Qdrant vector database will also be connected to the institutional data source, with an automated synchronization mechanism configured to periodically update its knowledge base with the latest authorized institutional documents and records (Tiwari et al., 2026).

Once the components have been successfully integrated, the GABAY will be deployed on a secure hosting infrastructure and configured for access by its intended users (e.g., administrators, faculty and staff, and students). Appropriate authentication, role-based access, and data security mechanisms will be applied to control access to institutional information. The deployed system will then serve as the working version of GABAY for the succeeding iterative testing and refinement phase.

Iterative Testing and Refinement. This phase will focus on systematically testing the integrated GABAY to identify functional issues, retrieval errors, and inconsistencies in AI-generated responses under realistic user scenarios (Baswardono et al., 2021).

Following system integration and deployment, the researchers will conduct scenario-based and multi-step testing using representative queries related to academic, administrative, and student support processes. The tests will examine whether user queries are correctly routed to the appropriate specialized agents, relevant institutional information is accurately retrieved, and responses are properly generated based on the retrieved data. More complex and unusual queries will also be tested to identify potential failures in agent routing, information retrieval, and system processing (Baswardono et al., 2021).

Based on the identified issues and system execution logs, the researchers will iteratively refine the agent prompts, routing rules, retrieval parameters, database lookup configurations, and system guardrails. Errors involving incomplete or unavailable institutional records will also be examined to improve how the system handles insufficient information and prevents unsupported responses. The testing–refinement cycle will be repeated until the identified functional and AI-related issues have been resolved and the system is ready for the succeeding multidimensional evaluation (Nechepurenko & Shuvalov, 2026).

Multidimensional Evaluation. This phase will assess the developed GABAY from both technical performance and user acceptance perspectives (Nguyen-Duc et al., 2025).

For the technical evaluation, the researchers will conduct empirical testing using standardized test questions covering the major functions and domains supported by GABAY. For the user evaluation, the researchers will administer an Extended Technology Acceptance Model (TAM) survey to the respondents. The detailed procedures, measures, instruments, and criteria for the technical and user evaluations will be discussed under Objectives No. 3 and 4, respectively.

Technical Performance Evaluation of GABAY

Following the development of the GABAY Management System, The third objective will evaluate its technical performance using an automated benchmarking suite and empirical evaluation framework across thirty-two (32) standardized query scenarios. The researchers and technical experts will conduct the evaluation using system execution results, ground-truth institutional records, and automated LLM-as-a-Judge assessments (Baswardono et al., 2021; Es et al., 2024). Technical performance will be evaluated using four metrics: Retrieval Accuracy, Response Quality, Reasoning Capability, and Report Generation Accuracy. The Data Analysis section will present the specific measures and formulas for these metrics.

Retrieval Accuracy will assess GABAY's ability to retrieve relevant information from the Qdrant vector store and Likha ERP to support user queries (Asai et al., 2024; Oche et al., 2025). It will determine whether the system retrieves the appropriate institutional documents and records needed to generate grounded responses (Es et al., 2024).

Response Quality will evaluate the factual accuracy, relevance, and clarity of the responses generated by GABAY based on retrieved institutional information (Es et al., 2024; Xie et al., 2026). Particular attention will be given to whether the generated responses remain grounded in verified Likha ERP records and official institutional documents (Nguyen-Duc et al., 2025; Xie et al., 2026).

Moreover, Reasoning Capability will assess GABAY's ability to interpret user intent, route queries to the appropriate specialized agents, and perform the necessary sequence of actions to respond to complex queries (Nechepurenko & Shuvalov, 2026; Park

& Kwon, 2026). This will include evaluating the Coordinator Agent's routing and orchestration across the Academic, LMS, Research, Extension, QA, and SAS agents (Jiang et al., 2025; Park & Kwon, 2026).

Finally, the Report Generation Accuracy will assess the ability of GABAY to generate structured reports that accurately reflect the corresponding institutional database records (Aftab et al., 2019; Tiwari et al., 2026). The generated tables and summaries will be checked for data consistency and adherence to the required report structure (Nguyen-Duc et al., 2025).

Technology Acceptance of GABAY

For Objective No. 4, the technology acceptance of GABAY will be determined using an Extended Technology Acceptance Model (TAM) based on the original model of Davis (1989) and its later extensions (Venkatesh & Davis, 2000; Venkatesh & Bala, 2008). Since GABAY uses conversational AI and automated decision support, Trust in the AI System will be included as an additional construct (Pillai & Sivathanu, 2020; Shin, 2021). A 16-item questionnaire will be used (see Appendix A), with four items each for Perceived Usefulness, Perceived Ease of Use, Trust in the AI System, and Behavioral Intention to Use, rated using a 5-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree).

Perceived Usefulness refers to the degree to which users believe that GABAY improves the efficiency and effectiveness of their academic and administrative tasks. Perceived Ease of Use refers to the degree to which users perceive GABAY as easy to learn, understand, and operate. Trust in the AI System refers to the degree to which users have confidence in GABAY's reliability, accuracy, and appropriate handling of information. Behavioral Intention to Use refers to the users' willingness and intention to

adopt and continue using GABAY for academic and administrative activities (Kasilingam, 2020).

To ensure that the technology acceptance of GABAY will be evaluated by its intended users, appropriate respondents will be selected from the different user groups of the system. The respondents will consist of administrators, faculty and staff, and students of DMMMSU-SLUC CCS. Stratified random sampling will be used for students and faculty members, while total enumeration will be applied to the two non-teaching staff and two administrators.

Based on the identified population, the study will have a total population (N) of 896, consisting of 862 students, 30 faculty members, two non-teaching staff, and two administrators. Using Slovin's Formula with a 5% margin of error ($e = 0.05$), the required overall sample size (n) was computed as follows:

$$n = \frac{N}{1 + N(e)^2} = \frac{896}{1 + 896(0.05)^2} = \frac{896}{1 + 896(0.0025)} = \frac{896}{3.24} \\ \approx 277 \text{ respondents}$$

The number of respondents sampled from each stratum n_k was allocated proportionally relative to each stratum's total representation in the overall population, calculated as:

$$n_k = \left(\frac{N_k}{N} \right) \times n$$

Table 1 summarizes the stratified distribution of respondent samples across all user categories.

Table 1. Stratified Distribution of Respondent Samples for Extended TAM Evaluation

Stratum (User Category)	Total Population	Sample Size	Percentage of Sample (%)	Sampling Method
Stratum 1: 1 st Year Students	223	69	24.8%	Proportional Random
Stratum 2: 2 nd Year Students	234	72	25.9%	Proportional Random
Stratum 3: 3 rd Year Students	213	65	23.4%	Proportional Random
Stratum 4: 4 th Year Students	192	59	21.2%	Proportional Random
Stratum 5: Faculty Members	30	9	3.2%	Proportional Random
Stratum 6: Non-Teaching Staff	2	2	0.7%	Total Enumeration (100%)
Stratum 7: Administrators	2	2	0.7%	Total Enumeration (100%)
Total Population / Sample	896	278	100.0%	Stratified Sampling / Total Enumeration

Data Gathered

The following data will be gathered during the development and evaluation of GABAY to support the objectives of the study. All data collected from administrators, faculty and staff, and students, as well as institutional records, will be handled in accordance with Republic Act No. 10173, or the Data Privacy Act of 2012.

Interview

The researchers will conduct semi-structured interviews with administrators, faculty and staff, and students to gather information about existing processes, user needs, challenges, and system requirements. An interview guide will be used to ensure consistency in the data-gathering process.

Document Analysis

Relevant institutional documents, including policy manuals, student handbooks, academic program curricula, and Likha ERP database schemas, will be collected and analyzed to identify information and requirements needed for the development and knowledge base of GABAY.

Benchmark Testing and System Logging

The researchers will conduct benchmark testing and automated system logging to gather technical performance data for GABAY. Data from the 32 standardized test queries will be collected to evaluate retrieval accuracy, response quality, reasoning capability, and report generation accuracy.

Alpha and Beta Testing

During alpha testing, system logs, test results, response times, and identified errors will be gathered to detect and correct technical issues before deployment. During beta testing, data will be gathered from end-users as they interact with the deployed GABAY system in actual academic and administrative tasks.

Survey Questionnaire

A survey questionnaire based on the Extended Technology Acceptance Model (TAM) will be administered to the selected respondents to gather data on Perceived

Usefulness, Perceived Ease of Use, Trust in the AI System, and Behavioral Intention to Use.

Data Analysis

To draw meaningful and empirically grounded conclusions from the gathered data, appropriate quantitative analyses, automated evaluation methods, and descriptive statistical treatments will be applied. The data analysis will be organized into two major sections: Technical Performance Analysis, which will evaluate Objective No. 3 in terms of retrieval accuracy, response quality, reasoning capability, and report generation accuracy; and Technology Acceptance Analysis, which will evaluate Objective No. 4 using the Extended Technology Acceptance Model (TAM).

Technical Performance Evaluation of GABAY

The technical performance of GABAY will be evaluated using thirty-two (32) standardized benchmark test queries designed to assess the system across different types and levels of complexity. The empirical data generated from these queries, including retrieved contexts, execution telemetry logs, multi-agent routing traces, tool execution records, generated responses, and generated reports, will be analyzed using Ragas-based evaluation, LLM-as-a-Judge scoring, and deterministic ground-truth matching, as appropriate to each performance measure. The evaluation will focus on four areas: retrieval accuracy, response quality, reasoning capability, and report generation accuracy.

To provide an overview of how each technical performance area will be assessed, Table 2 presents the corresponding evaluation metrics and the purpose of each measure.

Table 2. GABAY Technical Performance Evaluation

Performance Area	Evaluation Metrics	Purpose of Evaluation
Retrieval Accuracy	Context Precision; Context Recall; Tool Call Hit Rate	Determines whether GABAY retrieves relevant information and correctly accesses the required data source or API tool.
Response Quality	Faithfulness Score; Answer Relevancy	Determines whether generated responses are supported by the retrieved context and relevant to the user's query.
Reasoning Capability	Agent Routing Accuracy; Multi-Step Pass Rate	Determines whether queries are routed to the appropriate specialized agent and whether required sequential tool operations are successfully executed.
Report Generation Accuracy	Data Consistency Rate; Schema Adherence	Determines whether generated reports contain accurate data and conform to the required structure and format.

Benchmark Query Distribution. The benchmark queries are structured to represent varying levels of task complexity encountered when using GABAY. The 32 queries are classified into four complexity tiers: Easy (Level 1), Moderate (Level 2), Complex (Level 3), and Out-of-Scope (Level 4). This distribution allows the system to be evaluated not only on routine informational retrieval but also on operational tasks requiring multi-step reasoning, cross-domain agent coordination, and tabular report aggregation. Out-of-scope queries are included to determine whether GABAY can appropriately recognize and refuse requests beyond its intended functional boundaries, verifying traffic-light governance and policy adherence.

Table 3. Distribution of Benchmark Test Queries Across Sub-Objectives and Complexity

Levels					
Performance Sub-Objective	Easy (Level 1)	Moderate (Level 2)	Complex (Level 3)	Out-of-Scope (Level 4)	Total Queries
Retrieval Accuracy	2	3	2	1	8
Response Quality	2	3	2	1	8
Reasoning Capability	1	2	4	1	8
Report Generation Accuracy	2	3	3	0	8
Total	7	11	11	3	32

The selection of a focused 32 benchmark queries follows standard research practices for retrieval and multi-agent systems. In systems that query structured database APIs, a compact suite of hand-verified questions is significantly more reliable for catching errors than large, noisy test sets (Chen et al., 2024; Es et al., 2024). Testing complex multi-agent workflows requires a thorough, step-by-step review of how each agent reasons, selects tools, and routes information, making depth of analysis far more valuable than sheer query volume (Mialon et al., 2023; Pookduang & Chansanam, 2026).

Table 3 presents the distribution of the 32 benchmark test queries across the four technical performance areas and their corresponding complexity levels.

The data obtained from the benchmark testing will then be analyzed using the specific metrics identified in Table 2. Retrieval Accuracy will be analyzed using Context Precision, Context Recall, and Tool Call Hit Rate; Response Quality using Faithfulness and Answer Relevancy; Reasoning Capability using Agent Routing Accuracy and Multi-

Step Pass Rate; and Report Generation Accuracy using Data Consistency Rate and Schema Adherence. The computation and interpretation of each metric are presented in the succeeding subsections.

Retrieval Accuracy. The data for Retrieval Accuracy will be analyzed using the three metrics identified in Table 8: Tool Call Hit Rate, Context Precision, and Context Recall.

The Tool Call Hit Rate quantifies the proportion of benchmark queries that correctly trigger the target Likha ERP Directus/REST API tool with the required parameters (Pookduang & Chansanam, 2026; Tiwari et al., 2026).

The researchers will compute the Tool Call Hit Rate by dividing the number of benchmark queries that correctly invoke the required API tool with valid parameters by the total number of benchmark queries requiring tool execution, and multiplying the result by 100.

$$\text{Tool Call Hit Rate} = \left(\frac{T_{correct}}{T_{total}} \right) \times 100\%$$

For the Tool Call Hit Rate, an operational threshold of $\geq 90.0\%$ is established as Optimal (Production Ready), reflecting the benchmark standard for reliable API-mediated tool execution in enterprise RAG systems (Pookduang & Chansanam, 2026; Tiwari et al., 2026). Scores between 80.0% and 89.9% represent Acceptable performance with minor parameter formatting discrepancies, while scores from 70.0% to 79.9% indicate Moderate performance that requires schema refinement (Wang et al., 2024). Scores below 70.0% are

considered Unacceptable, indicating critical failure in invoking backend Directus or REST API endpoints and blocking system release (Pookduang & Chansanam, 2026).

In addition, context precision will measure whether the relevant context chunks retrieved from the Qdrant Vector Store are ranked higher than irrelevant chunks in the retrieved context (Es et al., 2024; Pookduang & Chansanam, 2026).

The researchers will compute Context Precision using the Ragas framework by evaluating the relevance of each retrieved context chunk according to its rank. A relevance value of 1 will be assigned if a retrieved chunk is relevant to the benchmark query, while 0 will be assigned if it is irrelevant. The precision at each rank will then be weighted by its relevance value and summed across the top- K retrieved chunks. The resulting value will be divided by the total number of relevant chunks within the top- K retrieved set.

$$\text{Context Precision@}K = \frac{\sum_{k=1}^K (\text{Precision@}k \times v_k)}{\text{Total Relevant Items in Top } K}$$

Where,

K = number of retrieved context chunks evaluated;

$\text{Precision@}k$ = precision calculated up to rank k

v_k = relevance indicator, where $v_k = 1$ if the context chunk at rank k is relevant and

$v_k = 0$ if it is irrelevant.

For Context Precision, a score range of 0.90 to 1.00 is defined as Optimal, indicating that relevant institutional document chunks from Qdrant are placed at the top ranks of the context window to minimize attention distraction in the LLM (Es et al., 2024; Saad-Falcon et al., 2024). Scores between 0.80 and 0.89 denote Acceptable ranking quality with slight contextual noise, whereas scores from 0.70 to 0.79 reflect

Moderate performance needing optimization (Chen et al., 2024). Scores falling below 0.70 represent Poor retrieval precision, triggering a restructuring of the Recursive Character Text Splitter parameters (250/50) and stricter metadata domain filters (Oche et al., 2025).

Finally, Context Recall will evaluate whether the Qdrant Vector Store and Likha ERP connectors successfully retrieve all the necessary reference statements required to answer a benchmark query (Es et al., 2024; Oche et al., 2025). The researchers will compute Context Recall by determining the number of ground-truth reference statements found within the retrieved context and dividing it by the total number of reference statements defined in the human-verified ground truth:

$$\text{Context Recall} = \frac{|\text{Retrieved Ground Truth Statements}|}{|\text{Total Ground Truth Statements}|}$$

Where,

Retrieved Ground Truth Statements = number of ground-truth reference statements found in the retrieved context

Total Ground Truth Statements = total number of reference statements required to correctly answer the benchmark query.

For Context Recall, a score of 0.90 to 1.00 is designated as Optimal, verifying that the retrieval pipeline extracts all ground-truth reference statements necessary to answer complex institutional queries (Es et al., 2024; Oche et al., 2025). Scores between 0.80 and 0.89 are classified as Acceptable, where most essential information is captured with minor omissions (Chen et al., 2024). Scores ranging from 0.70 to 0.79 indicate Moderate recall

requiring top- K expansion, while scores below 0.70 represent Unacceptable information loss that directly compromises the completeness of generated responses (Es et al., 2024; Pookduang & Chansanam, 2026)

Response Quality. This will determine how effectively GABAY generates responses that are supported by the retrieved institutional information and relevant to the user's query. It will be analyzed using two metrics: Faithfulness and Answer Relevancy (Es et al., 2024; Xie et al., 2026).

Faithfulness will measure the extent to which the factual claims in GABAY's generated response are supported by the retrieved context, thereby determining whether the response contains information that cannot be inferred from the retrieved institutional data (Chen et al., 2024; Nguyen-Duc et al., 2025; Xie et al., 2026).

The researchers will compute the Faithfulness Score by breaking down the generated response into individual factual claims and determining whether each claim is supported by the retrieved context. The number of context-supported claims will then be divided by the total number of factual claims made in the generated response:

$$\text{Faithfulness Score} = \frac{|\text{Number of Context – Supported Claims in Output}|}{|\text{Total Number of Claims Made in Output}|}$$

Where,

Context – Supported Claims in Output = number of factual claims in the generated response that are supported by the retrieved context; and

Total Number of Claims Made in Output = total number of factual claims identified in the generated response.

For the Faithfulness Score, a strict threshold of 0.90 to 1.00 is established as Optimal, confirming that all factual claims generated by GABAY are derived strictly from retrieved institutional evidence without hallucination (Chen et al., 2024; Es et al., 2024; Nguyen-Duc et al., 2025). Scores from 0.80 to 0.89 represent an Acceptable tier characterized by minor stylistic or inferential phrasing variations that do not alter factual meaning (Xie et al., 2026). Scores between 0.70 and 0.79 denote Moderate reliability requiring prompt guardrail adjustments, whereas scores below 0.70 are deemed Unacceptable, triggering an immediate deployment block due to high hallucination risk in official college policies (Chen et al., 2024; Pookduang & Chansanam, 2026).

Furthermore, answer relevancy will measure how well GABAY's generated response addresses the user's query without including unnecessary or unrelated information (Es et al., 2024).

The researchers will compute Answer Relevancy using the Ragas framework by generating m candidate questions from GABAY's generated response. The semantic cosine similarity between the embedding vector of the original user query and the embedding vector of each generated candidate question will then be computed. The resulting similarity scores will be averaged to determine the Answer Relevancy score:

$$\text{Answer Relevancy} = \frac{1}{m} \sum_{i=1}^m \text{CosineSimilarity}(e_q, e_{q_i})$$

Where,

m = number of candidate questions generated from GABAY's response;

e_q = embedding vector of the original user query

e_{qi} = embedding vector of the i^{th} candidate question generated from GABAY's response.

For Answer Relevancy, a score of 0.90 to 1.00 indicates Optimal performance, demonstrating that the generated response directly answers the user's intent without redundant, off-topic, or incomplete narrative elements (Es et al., 2024; Saad-Falcon et al., 2024). Scores between 0.80 and 0.89 represent Acceptable response alignment suitable for general academic assistance (Nguyen-Duc et al., 2025). Scores from 0.70 to 0.79 indicate Moderate relevancy with noticeable verbosity or topic drift, while scores below 0.70 represent Poor generation quality requiring prompt template revision (Es et al., 2024).

Reasoning Capability. Reasoning Capability will determine how effectively GABAY interprets user queries, routes them to the appropriate specialized agent, and coordinates multi-step tool execution when a query requires information or operations across multiple institutional domains. It will be analyzed using two metrics: Agent Routing Accuracy and Multi-Step Tool Execution Pass Rate (Nechepurenko & Shuvalov, 2026; Park & Kwon, 2026; Silvestri et al., 2024).

Agent Routing Accuracy will measure how effectively the Coordinator Agent routes a user query to the appropriate specialized agent based on the institutional domain required to answer the query. The specialized agents include the Curriculum and Instruction agent, Research agent, Extension agent, Quality Assurance agent, Faculty agent, Student Support agent, and Facilities agent.

The researchers will compute Agent Routing Accuracy by taking the number of queries successfully delegated to the correct specialist agent ($R_{correct}$), dividing it by the total number of test queries evaluated (R_{total}), and multiplying the quotient by 100 percent:

$$Agent\ Routing\ Accuracy = \left(\frac{R_{correct}}{R_{total}} \right) \times 100\%$$

Where,

$R_{correct}$ = number of benchmark queries correctly routed to the expected specialized agent

R_{total} = total number of benchmark queries evaluated for agent routing

For Agent Routing Accuracy, a performance level of $\geq 90.0\%$ is defined as Optimal, ensuring that the high-level Coordinator Agent reliably classifies user intent and delegates tasks to the correct domain specialist without routing errors (Nechepurenko & Shuvalov, 2026; Park & Kwon, 2026). Scores of 80.0% to 89.9% reflect Acceptable routing reliability, whereas scores from 70.0% to 79.9% indicate Moderate performance with boundary confusion between overlapping domains (e.g., Academic vs. QA) (Xie et al., 2026). Scores below 70.0% are classified as Unacceptable, as routing failures lead to cascading multi-agent communication breakdowns that require restructuring intent-classification prompts (Nechepurenko & Shuvalov, 2026).

Moreover, Multi-Step Tool Execution Pass Rate will measure how effectively GABAY completes queries that require a sequence of dependent tool executions across one or more specialized agents without missing required steps, losing parameter values, or encountering execution loops (Jiang et al., 2025; Pookduang & Chansanam, 2026).

The researchers will compute the Multi-Step Pass Rate by taking the total count of multi-step test queries where all sequential tool dependencies executed deterministically to completion (S_{passed}), dividing it by the total number of multi-step benchmark scenarios (S_{total}), and multiplying the result by 100 percent:

$$Multi - Step Tool Execution Pass Rate = \left(\frac{S_{passed}}{S_{total}} \right) \times 100\%$$

Where,

S_{passed} = number of multi-step benchmark queries in which all required tool execution steps are successfully completed

S_{total} = total number of multi-step benchmark queries evaluated

For the Multi-Step Tool Execution Pass Rate, a threshold of $\geq 90.0\%$ is established as Optimal, demonstrating that complex queries requiring sequential tool invocations execute deterministically across domain agents without execution loops or dropped parameters (Jiang et al., 2025; Pookduang & Chansanam, 2026). Pass rates between 80.0% and 89.9% represent Acceptable multi-step orchestration (Mialon et al., 2023). Scores from 70.0% to 79.9% represent Moderate performance requiring workflow timeout and retry adjustments, while pass rates below 70.0% are Unacceptable, signaling broken agentic control flows that halt production deployment (Pookduang & Chansanam, 2026).

Report Generation Accuracy. Report Generation Accuracy will determine how effectively GABAY generates accurate and properly structured tabular reports from institutional data. It will be analyzed using two metrics: Data Consistency Rate and Schema and Formatting Adherence (Aftab et al., 2019; Tiwari et al., 2026).

Data Consistency Rate will measure the accuracy of the numerical and textual values presented in GABAY-generated tabular reports by comparing them with the corresponding values from the ground-truth database query results (Nguyen-Duc et al., 2025; Pookduang & Chansanam, 2026).

The researchers will calculate the Data Consistency Rate by comparing each cell in the generated report against the corresponding cell from the direct SQL database query, dividing the total count of perfectly matched cells ($C_{matching}$) by the total number of expected cells in the ground-truth table (C_{total}), and multiplying the quotient by 100 percent:

$$Data\ Consistency\ Rate = \left(\frac{C_{matching}}{C_{total}} \right) \times 100\%$$

Where,

$C_{matching}$ = number of cells in the generated report that exactly match the corresponding cells in the ground-truth table

C_{total} = total number of expected cells in the ground-truth table

For the Data Consistency Rate, an evaluative threshold of $\geq 90.0\%$ is set as Optimal, verifying that numerical and textual cells in GABAY-generated tabular reports match direct SQL database queries cell-for-cell (Pookduang & Chansanam, 2026; Tiwari et al., 2026). Consistency scores from 80.0% to 89.9% represent an Acceptable standard for preliminary administrative summaries (Wang et al., 2024). Scores between 70.0% and 79.9% reflect Moderate consistency requiring query aggregation adjustments, while scores below 70.0% denote Unacceptable reporting discrepancies that necessitate backend SQL

tool schema corrections before administrative use (Nguyen-Duc et al., 2025; Pookduang & Chansanam, 2026).

In addition, Schema and Formatting Adherence will measure how effectively GABAY-generated tabular reports comply with the predefined structural and formatting requirements, including the required table structure, column headers, data formats, and sort order.

The researchers will compute the Schema Adherence by determining the number of generated tabular reports that completely comply with all predefined schema and formatting requirements ($F_{compliant}$), dividing it by the total number of generated tabular reports evaluated (F_{total}), and multiplying the quotient by 100 percent:

$$Schema\ Adherence\ Rate = \left(\frac{F_{compliant}}{F_{total}} \right) \times 100\%$$

Where,

$F_{compliant}$ = number of generated tabular reports that completely comply with all predefined schema and formatting requirements

F_{total} = total number of generated tabular reports evaluated

For Schema and Formatting Adherence, a rate of $\geq 90.0\%$ is defined as Optimal, validating that generated tabular reports fully comply with expected column structures, header conventions, data types, and sorting rules (Aftab et al., 2019; Tiwari et al., 2026). Adherence rates between 80.0% and 89.9% represent an Acceptable level where minor layout variations occur without corrupting the table schema (Nguyen-Duc et al., 2025). Rates from 70.0% to 79.9% indicate Moderate adherence requiring Pydantic schema

validation updates, while rates below 70.0% represent Unacceptable formatting failures that prevent frontend UI dashboard rendering (Aftab et al., 2019).

Technology Acceptance of GABAY

Quantitative survey responses gathered from the $n = 277$ stratified campus evaluators (Students across four-year levels, Faculty Members, and Administrators) will be analyzed using descriptive statistics, specifically the Weighted Mean ($\bar{x}$) and Standard Deviation (SD) (Atmowardoyo, 2018; Davis, 1989; Sahin & Mete, 2021; Venkatesh & Bala, 2008).

The Weighted Mean will be calculated for each of the sixteen (16) questionnaire items as well as the composite mean for each of the four constructs: perceived usefulness, perceived ease of use, trust in the AI system, and behavioral intention to use

The researchers will compute the Weighted Mean by multiplying the frequency count of responses (f) for each Likert scale option by its assigned weight (x , ranging from 1 to 5), calculating the summation of these products across all response levels ($\sum(f \cdot x)$), and dividing the resulting total by the total number of respondents (n) within that stratum or overall sample:

$$\bar{x} = \frac{\sum(f \cdot x)}{n}$$

Where:

$\bar{x}$ = Computed Weighted Mean score

f = Frequency of responses for a specific Likert rating scale

x = Assigned numerical weight of the Likert rating (1 to 5)

n = Total number of respondent evaluators within the stratum or overall sample

Standard Deviation (SD). The Standard Deviation will be computed alongside each mean score to measure the dispersion and consensus of evaluator ratings around the calculated mean.

The researchers will determine the Standard Deviation by subtracting the computed sample mean ($\bar{x}$) from each individual respondent's score (x_i), squaring each deviation value, summing these squared differences across all respondents ($\sum(x_i - \bar{x})^2$), dividing the total sum by the sample size minus one ($n - 1$) to obtain the sample variance, and calculating the principal square root of the result:

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Where:

x_i = Individual respondent rating

$\bar{x}$ = Computed sample mean score

n = Total number of respondents in the sample/stratum

Statistical Interpretation Scale. To provide qualitative meaning to the calculated weighted mean scores, an established 5-point Likert continuum scale will be utilized (Davis, 1989; Sahin & Mete, 2021; Venkatesh & Bala, 2008), as presented in Table 4.

Table 4. Statistical Interpretation Scale for Extended TAM Survey

Mean Range	Likert Weight	Verbal Interpretation	Level of Technology Acceptance
4.21 – 5.00	5	Strongly Agree	Highest Acceptance
3.41 – 4.20	4	Agree	High Acceptance
2.61 – 3.40	3	Neutral	Moderate Acceptance
1.81 – 2.60	2	Disagree	Low Acceptance
1.00 – 1.80	1	Strongly Disagree	Lowest Acceptance

As shown in Table 4, the statistical interpretation continuum was established using standard psychometric class-interval calculations to derive five equal, mutually exclusive score ranges with an interval width of 0.80 (Davis, 1989; Sahin & Mete, 2021; Venkatesh & Bala, 2008). Mean scores ranging from 3.41 to 5.00 indicate High to Highest Acceptance, reflecting that stakeholders perceive GABAY as highly useful, easy to operate, trustworthy, and ready for institutional adoption. Conversely, scores falling at or below 3.40 represent moderate acceptance or operational resistance that necessitates further system refinement. This standardized continuum will serve as the evaluative benchmark for interpreting the composite construct scores and comparing perceptual variations across Students, Faculty Members, and Administrators.

Chapter 3

RESULTS AND DISCUSSION

Chapter 4

SUMMARY, CONCLUSION, AND RECOMMENDATION

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APPENDICES

APPENDIX A

Extended Technology Acceptance Model Survey Questionnaire

Extended Technology Acceptance Model (TAM) Survey Questionnaire

Name (Optional): _____ Date: _____

Respondent Stratum:

- | | | |
|---|---|---|
| ☐ 1 st Year Student | ☐ 2 nd Year Student | ☐ 3 rd Year Student |
| ☐ 4 th Year Student | ☐ Faculty Member | ☐ Non Teaching Staff |
| ☐ Administrator | | |

Direction: Please put a check mark (✓) under the rating that best describes the indicators using the following rating scales to determine the level of acceptance of the developed system.

Scale:

- | | | |
|---|---|-------------------|
| 5 | - | Strongly Agree |
| 4 | - | Agree |
| 3 | - | Neutral |
| 2 | - | Disagree |
| 1 | - | Strongly Disagree |



Item Dimension	Questions	5	4	3	2	1
a. Perceived Usefulness	1. Using GABAY enables me to accomplish academic, administrative, and advising inquiries more quickly than manual searches.					
	2. Using GABAY improves my efficiency in accessing official college policies, curricula, and operational records.					
	3. GABAY simplifies college workflows by reducing the effort needed to navigate multiple menus and physical handbooks.					
	4. Overall, I find GABAY useful in supporting my daily academic or administrative tasks within the College.					
b. Perceived Ease of Use	1. Interacting with GABAY's natural language chat interface is clear, intuitive, and easy to understand.					
	2. I find it easy to get GABAY to retrieve the exact institutional information or generate the specific reports I need.					

	3. Learning how to operate the GABAY platform and its dashboards requires minimal effort without technical training.						
	4. Overall, I find the GABAY system user-friendly, responsive, and easy to navigate.						
c. Trust in the AI System	1. I am confident that the responses provided by GABAY are factually grounded in official DMMMSU-SLUC documents.						
	2. I trust GABAY to provide reliable advice and reports without producing fabricated or misleading information (hallucinations).						
	3. I feel secure that GABAY protects confidential institutional data and complies with privacy mandates (RA 10173).						
	4. Overall, I trust GABAY as a dependable and transparent decision-support system for our College.						
d. Behavioral Intention to Use	1. I intend to use GABAY regularly whenever I need to check academic policies, prerequisites, or institutional data.						
	2. I predict that I will frequently utilize GABAY as an alternative to manual consultations and spreadsheet reports.						
	3. I would recommend GABAY to other students, faculty members, and administrative staff across the campus.						
	4. Overall, I plan to integrate GABAY into my routine academic or administrative workflows in the future.						

Comments and Suggestions:

THANK YOU!

Respondent's Signature

APPENDIX B

Request for Thesis Adviser



DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
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COLLEGE OF COMPUTER SCIENCE
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June 25, 2026

DR. CHARLIE S. MARZAN
 Dean, College of Computer Science
 DMMMSU-SLUC
 Agoo, La Union

Madam:

We, the undersigned fourth-year students in the Bachelor of Science in Computer Science program, are currently enrolled in Thesis Writing 1, respectfully request that you serve as our **Thesis Adviser** for our proposed undergraduate research.

We intend to conduct a study on the following proposed research topic:

- GABAY: A Hierarchical Multi-Agent Retrieval-Augmented Conversational AI System for College Management, Decision Support, and Learning Services

Considering your expertise and experience in Computer Science, we believe your guidance, knowledge, and insights will greatly contribute to the successful completion of our research.

We sincerely hope for your favorable consideration of this request. Thank you very much for your time and support.

Thank you very much.

Respectfully yours,

Jayvee G. Reyes
 Leader

Johren Mae B. Biscarra
 Member

Karylle Montemayor
 Member

Jamaica Fea C. Aquino
 Member

Rica Castronuevo
 Member

Approved:

DR. CHARLIE S. MARZAN
 Thesis Adviser



CURRICULUM VITAE

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Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2017 – 2023</i>	Sto. Tomas National High School Santo Tomas, La Union
<i>2012 – 2017</i>	Patac Elementary school Patac, Santo Tomas, La Union
<i>2011 – 2012</i>	Tagumpay Elementary school Montalban, Rodriguez, Rizal

Awards and Honors Received

<i>June 2026</i>	Certificate of Completion of On-the-Job-Training Zyberlab Solutions Inc. Nibaliw East, San Fabian, Pangasinan
<i>2016 – 2023</i>	Consistent Honor Student Santo Tomas, La Union

2023 – 2024

Dean's Lister

Don Mariano Marcos Memorial State University
 Consolacion, Agoo, La Union

Work Experiences

2024 – present

Web Developer
 Freelancing

2026 – present

Junior Developer
 Zyberlab IT Solutions
 San Fabian, Pangasinan

Seminars and Trainings Attended

May – June 2026

Zyberlab IT Solutions On-the-Job-Training
 DMMMSU SLUC BSCS OJT Facilitator
 San Fabian, Pangasinan

Special Skills and Talents

- Expert in programming languages (C#, PHP, JavaScript, SQL, HTML5, CSS3), backend (ASP.NET Core, Entity Framework, Web API), frontend (jQuery, Tailwind, Bootstrap, AJAX), databases (SQL Server, MySQL), and tools (Git, Excel data processing).
- Strong in communication, leadership, problem-solving, technical writing, and working under pressure.

Certified true and correct.

JAYVEE G. REYES

Signature over Printed Name

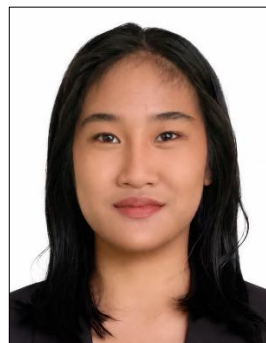
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Awards and Honors Received

<i>June 2026</i>	Certificate of Completion of On-the-Job-Training Department of Environment and Natural Resources DENR Compound, Gibraltar Rd, Baguio City
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Work Experiences

<i>May – June 2026</i>	Administrative Assistant Department of Environment and Natural Resources – Office of the Regional Executive Director
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Seminars and Trainings Attended

May – June 2026

Department of Environment and Natural Resources

On-the- Job-Training

DMMMSU SLUC BSCS OJT Facilitator

DENR Compound, Gibraltar Rd, Baguio City

Special Skills and Talents

- Proficient in Microsoft Word, with the ability to create, format, and edit professional documents, reports, and correspondence efficiently. Skilled in Microsoft Excel, including data entry, basic formulas, spreadsheet organisation, and creating clear, well-structured tables
- Strong technical aptitude and computer literacy, including the confident operation of various software applications and operating systems.

Certified true and correct.

JOHREN MAE B. BISCARRA

Signature over Printed Name

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<i>2011 – 2017</i>	Bail Elementary School Bail, Santo Tomas, La Union

Awards and Honors Received	
<i>June 2026</i>	Certificate of Completion of On-the-Job-Training Municipality of Agoo La Union Agoo, La Union

Work Experiences	
<i>May – June 2026</i>	Data Entry Clerk Municipality of Agoo La Union – Business Permits and Licensing Office

Seminars and Trainings Attended

May – June 2026

Municipality of Agoo On-the-Job-Training
DMMMSU SLUC BSCS OJT Facilitator
Agoo, La Union

Special Skills and Talents

- Strong organizational and time-management abilities, with a proven capacity to prioritize tasks and meet deadlines in busy environments.
- Effective multitasking ability, comfortably handling several responsibilities at once while maintaining quality and focus.

Certified true and correct.

KARYLLE P. MONTEMAYOR

Signature over Printed Name

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Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2017 – 2023</i>	San Fabian Integrated School SPED Center Inmalog Sur, San Fabian, Pangasinan
<i>2011 – 2017</i>	Inmalog Nortes Elementary School Inmalog Norte, San Fabian, Pangasinan

Awards and Honors Received

<i>June 2026</i>	Certificate of Completion of On-the-Job-Training Academia De Sta. Cecilia Foundation Inc. San Fabian, Pangasinan
------------------	--

Work Experiences

<i>2020</i>	Front Desk Associate PTA Beach Resort San Fabian, Pangasinan
<i>May – June 2026</i>	Administrative Assistant Academia De Sta. Cecilia Foundation Inc. – Registrar's Office

Seminars and Trainings Attended

May – June 2026

Academia De Sta. Cecilia Foundation Inc.
On-the-Job- Training
DMMMSU SLUC BSCS OJT Facilitator
San Fabian, Pangasinan

Special Skills and Talents

- Competent in managing digital files and folders across multiple platforms, ensuring documents are accurately stored and easily retrievable
- Ability to operate standard office systems and software efficiently while handling multiple tasks.

Certified true and correct.

RICA U. CASTRONUEVO

Signature over Printed Name

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Educational Background

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<i>2017 – 2023</i>	St. Mary of the Sea Academy Poblacion, Santo Tomas, La Union
<i>2011 – 2017</i>	Sto. Tomas Central School Poblacion, Santo Tomas, La Union

Awards and Honors Received

<i>June 2026</i>	Certificate of Completion of On-the-Job-Training Municipality of Santo Tomas, La Union Sto. Tomas, La Union
------------------	---

Work Experience

<i>May – June 2026</i>	Data Entry Clerk Municipality of Agoo La Union - Treasury Office
------------------------	---

Seminars and Trainings Attended

<i>May – June 2026</i>	Sto. Tomas LGU On-the-Job-Training DMMMSU SLUC BSCS OJT Facilitator Santo Tomas, La Union
------------------------	---

Special Skills and Talents

- Strong interpersonal skills, building positive working relationships with people
- Organised and able to manage time well when handling multiple responsibilities

Certified true and correct.

JAMAICA FEA C. AQUINO

Signature over Printed Name